Self Project Report

Communication Systems

Application of Communication System — LiFi

Design and Implementation of a Visible-Light Communication Link Using an Arduino Nano

LiFi (Light Fidelity) transmits data by modulating the intensity of an LED at high speed; an Arduino Nano drives the transmitter LED and decodes the signal received by a solar-panel photodetector, demonstrating visible-light communication as a practical alternative to RF-based wireless links.

Name	Abhishek
Roll Number	25M1079
Program	M.Tech, Communication Systems (EE1)

Indian Institute of Technology Bombay

(2025–2027)

Contents

List of Figures

Abstract

This project explores the application of LiFi (Light Fidelity) technology in modern communication systems as an innovative and high-speed alternative to traditional wireless communication methods. LiFi, which leverages visible light communication (VLC), transmits data through light-emitting diodes (LEDs), offering advantages such as enhanced security, reduced interference, and high data transmission rates. This technology presents an attractive solution to overcoming the limitations of radio frequency (RF) communication, particularly in environments where RF usage is restricted or limited.

The project aims to design and implement a functional LiFi system capable of transmitting data wirelessly using visible light. This includes a thorough investigation of LiFi components, operational principles, and data encoding techniques. By modulating the intensity of light from LEDs at very high speeds, the system enables data transfer in real time. The setup involves using an LED as the transmitter and a photodetector, such as a solar panel, as the receiver to decode the transmitted signals into usable data.

Through this project, we demonstrate the potential of LiFi as a robust communication solution with applications in various fields, including healthcare, aviation, and underwater communications, where radio frequencies face interference. Experimental results showcase the effectiveness of LiFi in transmitting data securely and efficiently, highlighting its relevance as a complementary technology to WiFi in future wireless communication infrastructures.

This study contributes to the growing body of research on LiFi technology and emphasizes its role in advancing communication systems by presenting a practical, low-cost, and high-speed data transmission model.

Chapter 1

Introduction to Communication System

1.1 Introduction

Communication may be defined as the transfer of information from one place to another via a sequence of procedures, or it can be defined as the communication of information itself. For communication to be effective, both the sender and the receiver must communicate in the same language.

Every living organism on the planet is confronted with the need to continually transmit and receive information from others in the surrounding environment. Throughout history, human beings have made an ongoing effort to increase the quality of communication with other human beings. From prehistoric to contemporary times, the methods and languages used in communication have continued to evolve and satisfy the increasing demands in terms of complexity and the speed of information transmission.

1.2 Types of Communication Systems

Depending on signal specification or technology, the communication system is classified as follows:

- **Analog** — Analog technology communicates data as electronic signals of varying frequency or amplitude. Broadcast and telephone transmission are common examples of analogue technology.

Figure 1.1: Analog v/s Digital Communication

- **Digital** — In digital technology, the data are generated and processed in two states: High (represented as 1) and low (represented as 0). Digital technology stores and transmits data in the form of 1s and 0s.

1.3 Basic Elements of a Communication System

There are three critical components to every communication system: the transmitter, the channel, and the receiver.

Information
Message or information is the entity that is to be transmitted. It can be in the form of audio, video, temperature, picture, pressure, etc.

Signal The single-valued function of time that carries the information. The information is converted into an electrical form for transmission.

Transducer
A device or an arrangement that converts one form of energy to another. An electrical transducer converts physical variables such as pressure, force, and temperature into corresponding electrical signal variations. For example, a microphone converts audio signals into electrical signals, and a photodetector converts light signals into electrical signals.

Amplifier
The electronic circuit or device that increases the amplitude or the strength of the transmitted signal is called an amplifier. When the signal strength becomes less than the required value, amplification can be done anywhere between the transmitter and receiver. A DC power source is provided for the amplification.

Modulator
As the original message signal cannot be transmitted over a large distance because of its low frequency and amplitude, it is superimposed with a high-frequency, high-amplitude carrier wave. This phenomenon of superimposing a message signal on a carrier wave is called modulation, and the resultant wave is a modulated wave which is to be transmitted. The common modulation types are:

(a) **Amplitude Modulation (AM)** — the amplitude of the carrier wave is varied in proportion to the message signal, keeping the carrier frequency constant.

(b) **Frequency Modulation (FM)** — the frequency of the carrier wave is varied with the message signal; it is more noise-resilient than AM.

(c) **Phase Modulation (PM)** — the phase of the carrier wave is varied with the message signal, giving good noise immunity.

Transmitter
The arrangement that processes the message signal into a suitable form for transmission and, subsequently, reception.

Antenna
A structure or device that radiates and receives electromagnetic waves, used in both transmitters and receivers. It is basically a metallic object, often a collection of wires; the electromagnetic waves are polarised according to the position of the antenna.

Channel
A physical medium such as wire, cable, or space through which the signal passes from the transmitter to the receiver. Noise, attenuation, and distortion are the major channel impairments.

Noise One of the channel imperfections in the received signal at the destination. External sources include interference from nearby transmitted signals (cross-talk), lightning, solar or cosmic radiation, and automobile-generated radiation; these can be minimised through appropriate channel design, shielding, and digital transmission. Internal sources include random electron motion and collision noise in conductors and thermal noise from diffusion/recombination of charge carriers, which can be minimised by cooling and using digital transmission.

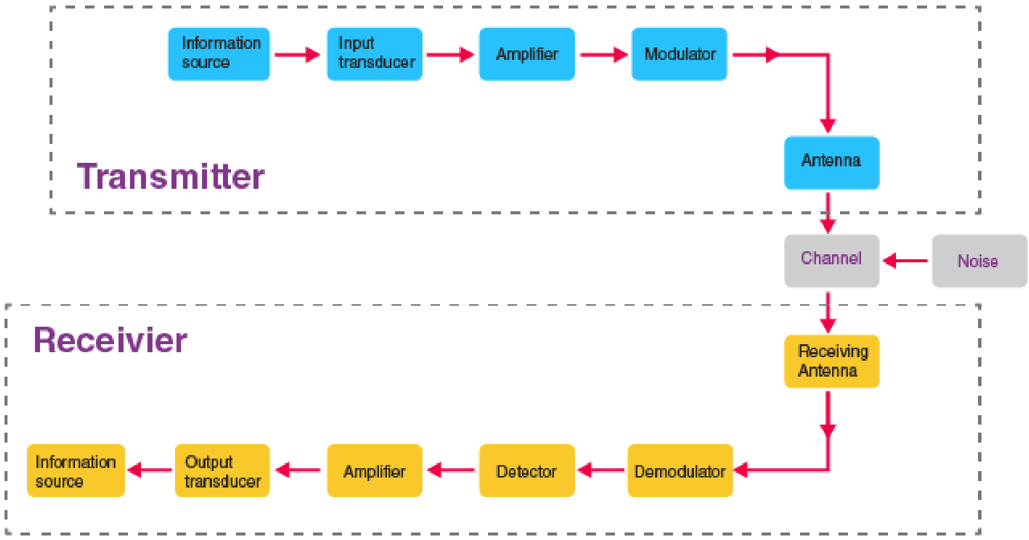

Figure 1.2: Block diagram of a communication system

1.4 Process of Communication

The information can be sent over a communication system from either a person or a machine. A transmitter is positioned at one location and a receiver is located at another, with the channel serving as the medium between the two. The message signal originates from the source and is fed into the transmitter.

The physical medium linking the transmitter and receiver (wired or wireless) is the channel. The broadcast signal may be warped due to channel imperfections, and noise may be added to the transmitted signal as it propagates.

Information is the concept or message being conveyed — also known as knowledge. A message may consist of a single message or a series of messages, in the form of a symbol, a code, a sequence of words, or any other predetermined unit. A transducer, modulator, amplifier and transmitting antenna are the transmitter components in radio transmission.

1.5 Terms Used in a Communication System

In an electronic communication system, electric circuits transmit, process, and receive data. Transmission happens through the transmitter, processing happens through the medium, and reception happens through the receiver. Data is transmitted in digital or analogue form. The basic terminology includes:

Communication medium

The channel used to transmit the signal from transmitter to receiver, e.g. copper wire and satellite systems.

Signal The information or data transmitted and received. It may be analogue (continuous variation, e.g. human voice) or digital (discrete 0/1 values).

Amplitude The signal height or strength; weak signals are strengthened through amplification.

Attenuation The process by which the amplitude of the signal decreases, particularly over long transmission distances.

Transducers Electrical devices that transform a physical quantity into an electrical quantity (and vice versa), consisting of sensing and transduction elements — for example, converting light or heat into an electrical signal.

Chapter 2

Introduction to LiFi

LiFi technology allows connection to the internet using light from lamps, streetlights, or LED televisions. In addition to being cheaper, safer, and faster than WiFi, it does not need a router — pointing a mobile device or tablet towards a light bulb is enough to surf the web.

2.1 What is LiFi Technology?

LiFi (Light Fidelity) is a bidirectional wireless system that transmits data via LED or infrared light. It was first unveiled in 2011 and, unlike WiFi (which uses radio frequency), LiFi technology only needs a light source with a chip to transmit an internet signal through light waves.

This is an extraordinary advance over today's wireless networks. LiFi multiplies the speed and bandwidth of WiFi, 3G, and 4G, which have limited capacity and become saturated as the number of users increases, causing crashes, reduced speeds, and interrupted connections.

With LiFi, however, a band frequency of 200,000 GHz — versus the maximum 5 GHz of WiFi — is 100 times faster and can transmit far more information per second. A 2017 study by the University of Eindhoven obtained a download rate of 42.8 Gbit/s with infrared light at a radius of 2.5 m, when the best WiFi would barely reach 300 Mbit/s.

Figure 2.1: Some basic uses of LiFi

2.2 LiFi v/s WiFi

LiFi technology is faster, cheaper, and more secure than WiFi. Its main advantages include:

Faster Current WiFi speed oscillates between 11 and 300 Mbit/s, while LiFi's most widely accepted speed is 10 Gbit/s, with demonstrated peaks up to 224 Gbit/s — fast enough to download a 1.5 Gbit film in thousandths of a second.

Cheaper and more sustainable
Up to 10 times cheaper than WiFi, requiring fewer components and less energy — all that is needed is to turn on a light.

More accessible
Any light fitting can be converted into an internet connection point by fitting a simple LiFi emitter.

More secure
Light does not pass through walls like radio waves do, preventing intruders from intercepting LiFi communications from outside the room.

More bandwidth
The light spectrum is 10,000 times wider than the radio spectrum, increasing the volume of data that can be carried per second.

More reliable
LiFi transmits its signal without interruptions, making communication more stable than WiFi.

No interference
Light does not interfere with radio communications or compromise transmissions from aircraft, ships, etc.

Wireless and invisible
LiFi dispenses with the router and can operate with infrared light (invisible to the human eye) or low-intensity visible LED light.

No saturation
Internet connection via light could prevent the collapse of the radio spectrum, which, according to LiFi's inventor Harald Haas, could otherwise occur by 2025.

With the emergence and development of LiFi technology, many foreshadow the obsolescence of WiFi and other wireless networks. It remains to be seen whether streetlights, in addition to illuminating our streets, will connect us to the internet at the speed of light.

2.3 Advantages of LiFi

1. **High Data Transfer Rates** — LiFi enables fast data transmission, making it suitable for internet applications.

2. **Enhanced Security** — LiFi relies on line-of-sight communication, so the signal is confined to a limited area and does not pass through walls, reducing the risk of unauthorized access.

3. **Energy Efficiency** — LiFi devices consume low power, making them ideal for IoT applications and reducing energy usage in lighting systems.

4. **Alleviates RF Spectrum Congestion** — Since LiFi operates in the optical spectrum, it helps reduce congestion in the crowded RF spectrum.

5. **Health Benefits** — Optical communication is safer than RF, as it poses no known health risks.

6. **Easy Installation** — Setting up a LiFi system is straightforward and offers potential energy savings in lighting systems.

2.4 Disadvantages of LiFi

1. **Limited Range** — LiFi works only within the range of the light source and cannot penetrate walls, limiting its range compared to WiFi.

2. **Outdoor Limitations** — Sunlight and other optical sources can interfere with LiFi, making it less effective outdoors.

3. **New Infrastructure Required** — Adopting LiFi requires new infrastructure, increasing the initial setup cost.

4. **Constant Light Requirement** — LiFi requires lights to remain on for continuous internet access, which can waste energy, especially at night.

5. **Limited Usability in Darkness** — LiFi systems cannot function in the dark, which is inconvenient for certain situations, such as using the internet before sleep.

Chapter 3

Components and Circuit Diagrams

3.1 Components Used

Arduino Nano

A compact, versatile microcontroller based on the ATmega328. It was chosen for this project due to its small size, ease of programming, and sufficient I/O pins to manage both the transmitter and receiver circuits. The Arduino Nano operates at 5 V and has 14 digital I/O pins, making it suitable for controlling LEDs and receiving signals from photodiodes or other light-sensitive sensors.

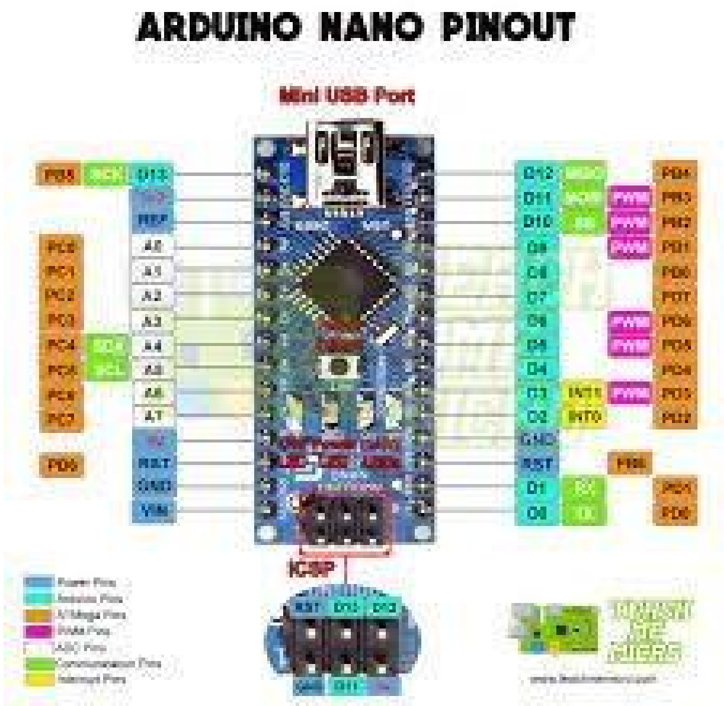

Figure 3.1: Arduino Nano pinout

12V LED

A 12V LED is a type of light-emitting diode designed to operate at a 12-volt power supply, commonly used in applications where higher voltage is required, such as automotive lighting, industrial lighting, and home lighting systems. Unlike standard LEDs, which typically operate at around 2–3 V, 12V LEDs are equipped with additional circuitry (usually a resistor) to regulate the higher input voltage.

Solar Panel

A solar panel is a device designed to convert sunlight into electrical energy through the photovoltaic (PV) effect. Solar panels consist of many solar cells, typically made of silicon, that absorb

Figure 3.2: 12V LED

photons from sunlight and release electrons, generating an electric current. In this project, the solar panel serves as the photodetector at the receiver.

Figure 3.3: Solar Panel

Arduino IDE

The Arduino Integrated Development Environment (IDE) is an open-source platform used for writing, compiling, and uploading code to Arduino boards. It provides a simple interface for developing and testing programs (sketches) that control the connected hardware.

Jumper Wires

Jumper wires are simple, insulated electrical wires used to make temporary or semi-permanent connections between components on a breadboard or other electronic circuits. They are essential for prototyping and are available as male-to-female, female-to-female, and male-to-male types.

Figure 3.4: Arduino IDE

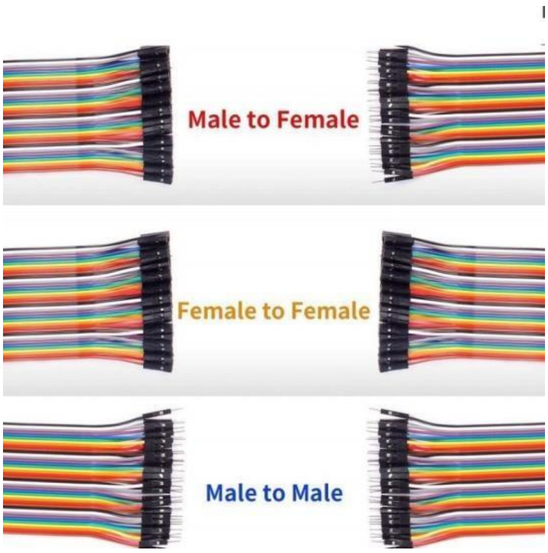

Figure 3.5: Jumper wires

IRLZ44N MOSFET

To drive the 12V LED circuit with both LiFi and the Arduino Nano, a transistor or MOSFET that can switch fast enough for optical data transmission is required. Switching speed is critical for reliable LiFi communication, so an IRLZ44N MOSFET was chosen over a regular transistor for its faster switching speed and better efficiency.

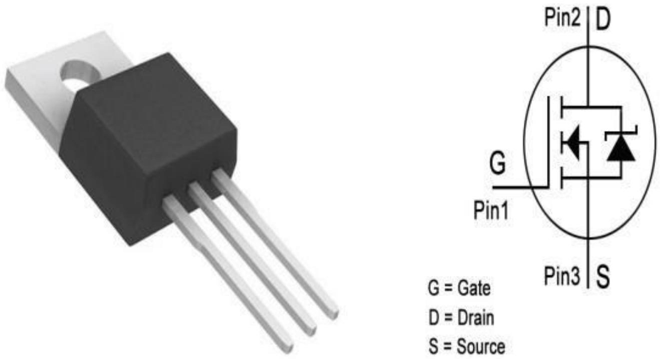

Figure 3.6: IRLZ44N MOSFET

Resistor

Resistors are fundamental electronic components that limit the flow of electric current in a circuit. They are essential for controlling voltage and current levels and are used for voltage division, current limiting, and signal attenuation.

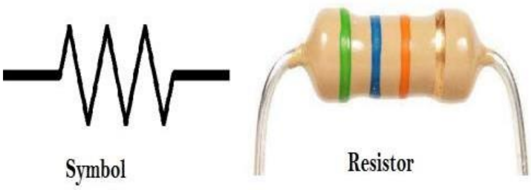

Figure 3.7: Resistor

3.2 Circuit Diagram

3.2.1 Transmitter Circuit Diagram

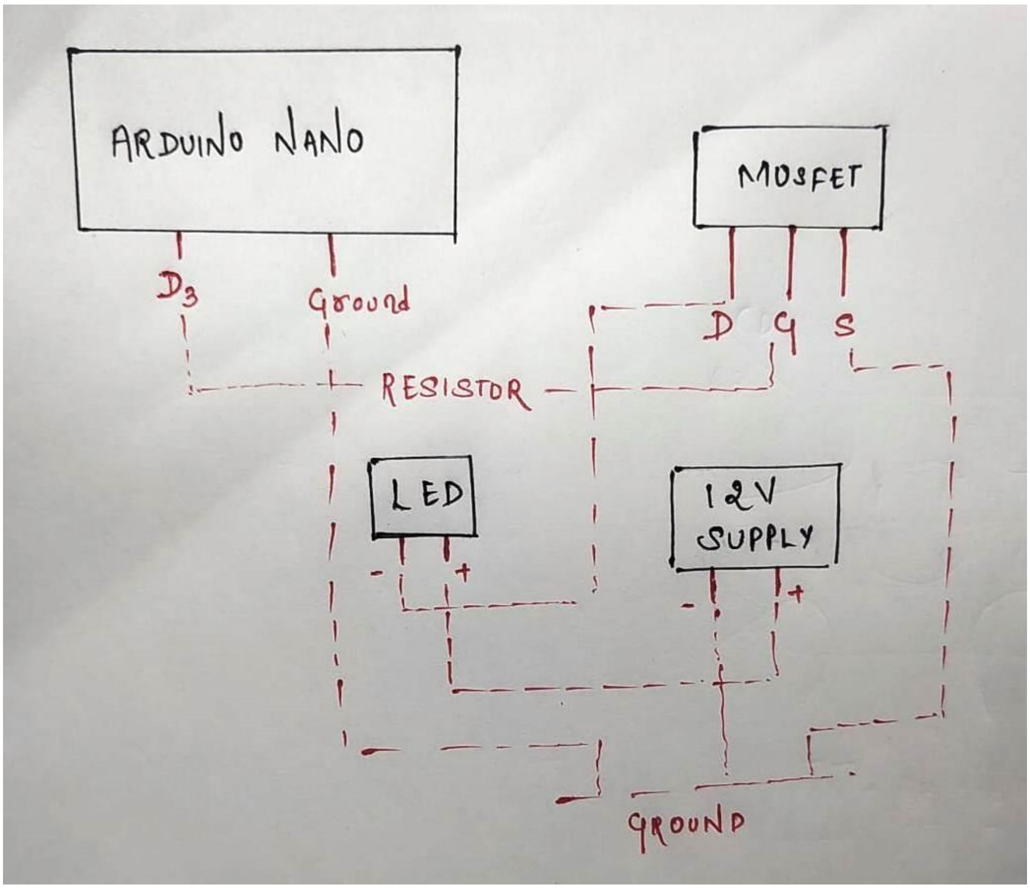

Figure 3.8: Transmitter circuit diagram

The Arduino Nano's digital output pin D3 drives the gate of the IRLZ44N MOSFET through a current-limiting resistor. The MOSFET switches the 12V-supplied LED on and off at high speed according to the digital data stream, with a common ground shared across the Arduino, MOSFET, and 12V supply.

3.2.2 Receiver Circuit Diagram

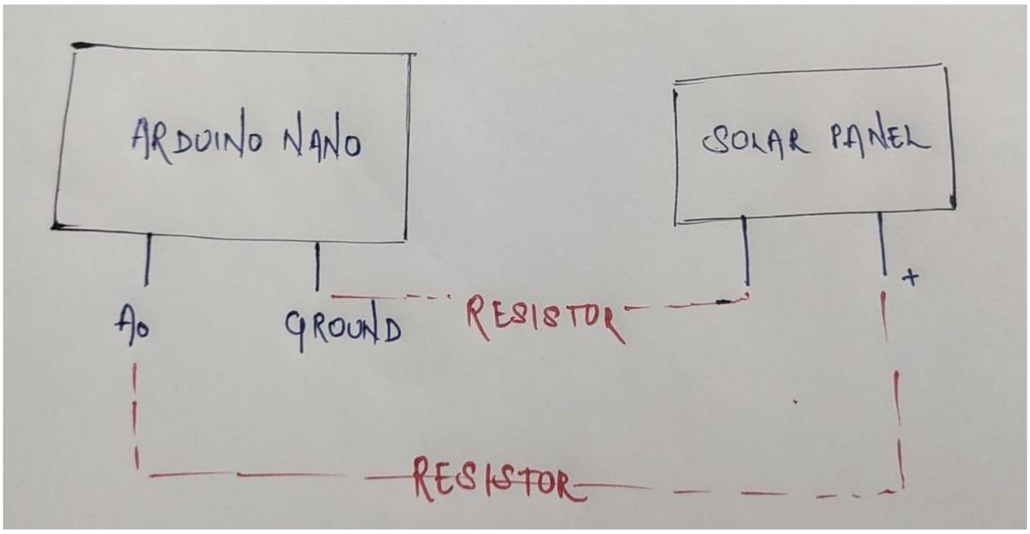

Figure 3.9: Receiver circuit diagram

The solar panel acts as the photodetector: light intensity variations from the transmitter LED generate a proportional voltage, which is fed (through a resistor divider) into the Arduino Nano's analog input pin A0 for sampling and decoding.

Chapter 4

Code Used for One-Way Communication

4.1 Transmitter Code

4.1.1 For Character Transmission

```
void setup() {
  Serial.begin(9600);
  pinMode(9, OUTPUT); // Laser module connected to pin 9
  Serial.println("Transmitter ready");
}

void loop() {
  if (Serial.available() > 0) {
    char data = Serial.read();
    Serial.print("Transmitting: ");
    Serial.println(data);
    sendChar(data);
  }
}

void sendChar(char c) {
  // Send start bit
  digitalWrite(9, HIGH);
  delay(200);

  // Send each bit of the character
  for (int i = 0; i < 8; i++) {
    bool bit = bitRead(c, i);
    digitalWrite(9, bit ? HIGH : LOW);
    delay(200); // Delay for each bit transmission
  }

  // Send stop bit
  digitalWrite(9, LOW);
  delay(200); // Short delay between characters
  Serial.println(" - Transmission complete");
}
```

Listing 4.1: Transmitter code — single-character transmission

4.1.2 For String Transmission

```
const int transmitPin = 9; // Digital pin to transmit data

void setup() {
  pinMode(transmitPin, OUTPUT);
  digitalWrite(transmitPin, LOW); // Ensure the line is low initially
  Serial.begin(9600); // Initialize serial communication for user input
  Serial.println("Transmitter ready.");
  Serial.println("Enter a string to send:");
}
```

```

void loop() {
  // Check if data is available from the Serial Monitor
  if (Serial.available() > 0) {
    String input = Serial.readStringUntil('\n'); // Read the input string
    Serial.print("Sending: ");
    Serial.println(input);
    sendString(input);
    Serial.println("Message sent. Enter another string to send:");
  }
}

void sendString(String str) {
  for (int i = 0; i < str.length(); i++) {
    sendChar(str[i]);
    delay(100); // Delay between characters (adjust as needed)
  }
}

void sendChar(char c) {
  // Start Bit
  digitalWrite(transmitPin, HIGH);
  delay(100); // Duration of the start bit
  Serial.print("Start Bit: 1 ");

  // Data Bits (7 bits, LSB first)
  for (int i = 0; i < 7; i++) {
    bool bit = bitRead(c, i);
    digitalWrite(transmitPin, bit ? HIGH : LOW);
    delay(100); // Duration of each data bit
    Serial.print(bit ? "1 " : "0 ");
  }

  // Stop Bit (optional, here we ensure the line is low after data bits)
  digitalWrite(transmitPin, LOW);
  delay(100); // Duration of the stop bit
  Serial.println("(Stop Bit: 0)");
}
```

Listing 4.2: Transmitter code — string transmission

4.2 Receiver Code

4.2.1 For Character

```
void setup() {
  Serial.begin(9600); // Start serial communication at 9600 baud
  Serial.println("Receiver ready. Waiting for data...");
}

void loop() {
  char c = receiveChar(); // Call the function to receive a character
  if (c != 0) { // Check if a valid character is received
    Serial.print("Received: ");
    Serial.println(c);
  }
  delay(200); // Delay for stability
}

char receiveChar() {
  char c = 0;
  bool startBitDetected = false;

  // Wait for the start bit
  while (!startBitDetected) {
    int sensorValue = analogRead(A0); // Read from the solar panel
    if (sensorValue > 550) { // Adjust threshold as necessary
      startBitDetected = true; // Start bit detected
    }
  }
  delay(200); // Wait to process the start bit

  Serial.print("Receiving bits: ");
  for (int i = 0; i < 20; i++) { // Read bits
    int sensorValue = analogRead(A0);
    bool bit = sensorValue > 550; // Check if it's a 1 or 0
    bitWrite(c, i, bit); // Store the bit in the character
    Serial.print(bit ? "1" : "0"); // Print the received bit
    delay(200); // Delay for each bit reception
  }

  // Wait for the stop bit
  delay(200);
  Serial.println(); // Print a new line for clarity
  return c; // Return the received character
}
```

Listing 4.3: Receiver code — single-character reception

4.2.2 For String

```
const int receivePin = A0; // Analog pin to receive data
const int threshold = 600; // Threshold to determine HIGH or LOW
String receivedString = ""; // Buffer to store the received string
unsigned long lastReceiveTime = 0;   // Timer to implement timeout
const unsigned long timeoutDuration = 5000; // 5 seconds timeout

void setup() {
  Serial.begin(115200); // Initialize serial communication for debugging
  Serial.println("Receiver ready. Waiting for data...");
}

void loop() {
  char receivedChar = receiveChar(); // Attempt to receive a character
  if (receivedChar != 0) { // If a valid character is received
    Serial.print("Received: ");
    Serial.println(receivedChar);
    receivedString += receivedChar; // Append to the buffer
    lastReceiveTime = millis(); // Reset the timeout timer

    // Check for the termination character (newline)
    if (receivedChar == '\n') {
      Serial.println("Transmission complete.");
      Serial.print("Full message: ");
      Serial.println(receivedString);
      receivedString = ""; // Reset the buffer for the next transmission
    }
  }

  // Reset the receiver if no data is received for a certain duration
  if (millis() - lastReceiveTime > timeoutDuration &&
      receivedString.length() > 0) {
    Serial.println("Transmission timeout. Resetting receiver.");
    receivedString = "";
    lastReceiveTime = millis(); // Reset the timer
  }

  delay(10); // Short delay for stability without blocking
}
```

Listing 4.4: Receiver code — string reception (Part 1)

```
char receiveChar() {
  char c = 0;
  bool startBitDetected = false;
  unsigned long startTime = millis(); // Start timeout timer

  // Wait for the start bit (HIGH signal) with a timeout
  while (!startBitDetected) {
    int sensorValue = analogRead(receivePin); // Read from the sensor
    if (sensorValue > threshold) { // Check if the signal is HIGH
      startBitDetected = true; // Start bit detected
    }
    // Check for timeout while waiting for start bit
    if (millis() - startTime > timeoutDuration) {
      return 0; // Return 0 indicating no character received
    }
    delay(5); // Short delay to prevent tight looping
  }

  delay(100); // Wait to align with the center of the first data bit

```

```
  // Read the next 7 data bits
  for (int i = 0; i < 7; i++) {
    int sensorValue = analogRead(receivePin);
    bool bit = sensorValue > threshold; // Determine if the bit is HIGH or LOW
    bitWrite(c, i, bit); // Store the bit in the character
    delay(100); // Duration of each data bit
  }

  return c; // Return the received character
}
```

Listing 4.5: Receiver code — string reception (Part 2, `receiveChar()`)

Chapter 5

Two-Way Communication

Two-way communication is an essential aspect of modern communication systems, including LiFi. It allows devices to both send and receive information, ensuring a dynamic and interactive data exchange. In the context of this LiFi project, implementing two-way communication opens up a wide range of possibilities for real-time applications, such as internet access, video conferencing, and IoT devices.

5.1 Need for Two-Way Communication in LiFi

1. **Interactive Data Exchange** — Two-way communication enables devices to request, confirm, and acknowledge data, improving reliability, and ensures seamless interaction between users and systems, similar to WiFi and other wireless technologies.

2. **Error Detection and Correction** — Bi-directional communication allows devices to detect transmission errors and request retransmission of corrupted data, ensuring data integrity.

3. **Control and Feedback Mechanism** — Control commands can be sent from the receiver back to the transmitter, enabling real-time adjustments for better performance.

4. **Practical Applications** — Use cases like live streaming, video calls, and interactive gaming rely on two-way communication for uninterrupted, responsive user experiences; IoT systems using LiFi need two-way communication to exchange commands and status updates.

5. **System Efficiency** — Two-way communication can optimize bandwidth utilization by allowing acknowledgments, prioritizing messages, and managing resources effectively.

Conclusion

This project demonstrated a functional one-way LiFi (visible-light communication) link built around an Arduino Nano, a 12V LED transmitter switched through an IRLZ44N MOSFET, and a solar-panel photodetector at the receiver. Character- and string-level data were encoded as start/data/stop bit sequences and transmitted by modulating the LED's light intensity, while the receiver sampled the solar panel's analog output to reconstruct the original message.

The implementation confirms the central claims explored in this report: LiFi can achieve secure, line-of-sight data transfer using low-cost, readily available components, while avoiding the electromagnetic interference and RF spectrum congestion associated with WiFi. At the same time, the practical limitations observed — range confined to the light source, sensitivity to ambient/outdoor light, and the requirement for the transmitter LED to remain on — align with the disadvantages identified in Chapter 2.

The extension to two-way communication (Chapter 5) outlines a clear path toward a more capable system supporting acknowledgements, error correction, and real-time feedback, which would be necessary for practical applications such as IoT device control or interactive data exchange. Overall, the project validates LiFi as a viable, complementary technology to WiFi for short-range, high-security, low-interference communication, and provides a working foundation for further development of higher-speed, bidirectional VLC systems.

References

1. Sudharshan B, Jahnavi A P, Jhanavi V, “Text Transmission using Li-Fi,” *International Advanced Research Journal in Science, Engineering and Technology*, Vol. 8, Issue 7, July 2021.

2. V. Parthasaradi, T. Senthil Kumar, R. Aswin Saravanan, A. Andro Edwin, M. Yohesh, S. Kamalesh, K.K. Shanthosh Shree, R. Deepa, “Text Data Transfer Using Li-Fi Communication,” *International Journal of Innovative Science and Research Technology*, Volume 8, Issue 1, January 2023.

3. ShabanaParveen M, Siddarthan K, Vignesh T, Ajay Krishnaa A R, “Data transmission using Li-Fi technology,” JETIR, June 2019, Volume 6, Issue 6.

4. Zahraa Tareq Aldarkazaly, Manal Fadhil Younus, Zainab Salim Alwan, “Data Transmission Using Li-Fi Technique,” *International Journal of Advanced Science and Technology*, Vol. 29, No. 03, (2020), pp. 7367–7382.